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Cathode: A thin plate of pure copper metal.

Anode: A block of impure copper metal.

Electrolyte: Copper sulphate solution acidified with sulphuric acid.

When electric current is passed through the electrolytic solution, pure copper gets deposited at the cathode and the impurities settle at the bottom of the anode in the form of sludge called anode mud.

Physical Properties of Copper

Copper is a reddish brown metal, with high lustre, high density and high melting point (1356°C).

Chemical Properties of Copper

i. **Action of Air and Moisture:** Copper gets covered with a green layer of basic copper carbonate in the presence of CO_2 and moisture.

$$2\,Cu + O_2 + CO_2 + H_2O \rightarrow CuCO_3.Cu(OH)_2$$

ii. **Action of Heat**: On heating at different temperatures in the presence of oxygen, copper forms two types of oxides CuO and Cu_2O.

$$2\,Cu + O_2 \xrightarrow{\text{below 1370K}} 2\,CuO$$

(copper II oxide- black)

$$4\,Cu + O_2 \xrightarrow{\text{above 1370K}} 2\,Cu_2O$$

(copper I oxide – red)

iii. **Action of Acids:**

a) With dilute HCl and dilute H_2SO_4:

Dilute acids such as HCl and H_2SO_4 have no action on these metals in the absence of air. Copper dissolves in these acids in the presence of air.

$$2\,Cu + 4\,HCl + O_2\,(air) \longrightarrow 2\,CuCl_2 + 2\,H_2O$$

b) With dil. HNO_3:

Copper reacts with dil. HNO_3 with the liberation of Nitric Oxide gas.

$$3\,Cu + 8\,HNO_3 \rightarrow 3\,Cu(NO_3)_2 + 2\,NO\uparrow + 4H_2O$$

iv) Action of Chlorine:

Chlorine reacts with copper, resulting in the formation of copper(II) chloride.

$$Cu + Cl_2 \longrightarrow CuCl_2$$

v) Action of Alkalis:

Copper is not attacked by alkalis.

Uses of Copper:

i. It is extensively used in manufacturing electric cables and other electric appliances.
ii. It is used for making utensils, containers, calorimeters and coins,
iii. It is used in electroplating.
iv. It is alloyed with gold and silver for making coins and jewels

8.9 METALLURGY OF IRON

Occurrence:

Iron is the second most abundant metal available next to aluminium. It occurs in nature as oxides, sulphides and carbonates. The ores of iron are as follows:

Ores of iron	Formula
Haematite	Fe_2O_3
Magnetite	Fe_3O_4
Iron pyrite	FeS_2

Iron is chiefly extracted from haematite ore (Fe_2O_3)

i. **Concentration by Gravity Separation:** The powdered ore is washed with a steam of water. As a result, the lighter sand particles and other impurities are washed away and the heavier ore particles settle down.

ii. **Roasting and Calcination:** The concentrated ore is strongly heated in a limited supply of air in a reverberatory furnace. As a result, moisture is driven out and sulphur, arsenic and phosphorus impurities are oxidized off.

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